



System Analysis and Design

CSE 307

Lecture 05

Information Gathering: Unobtrusive Methods



Learning Objectives

- Recognize the value of unobtrusive methods for information gathering.
- Understand the concept of sampling for human information requirements analysis.
- Construct useful samples of people, documents, and events for determining human information requirements.
- Create an analyst's playscript to observe decision-maker activities.
- Apply the STROBE technique to observe and interpret the decision-maker's environment.



Unobtrusive Methods

- Less disruptive
- Insufficient when used alone
- Multiple methods approach
- Used in conjunction with interactive methods



Major Topics

- Sampling
- Quantitative document analysis
- Qualitative document analysis
- Observation
- STROBE
- Applying STROBE



Sampling

- A process of systematically selecting representative elements of a population
- Involves two key decisions:
 - What to examine
 - Which people to consider



Need for Sampling

The reasons systems analysts do sampling are:

- Containing costs
- Speeding up the data gathering
- Improving effectiveness
- Reducing bias



Need for Sampling

- Too costly to:
 - Examine every scrap of paper
 - Talk with everyone
 - Read every web page from the organization



Sampling

- Sampling helps accelerate the process by gathering selected data rather than all data for the entire population
- The systems analyst is spared the burden of analyzing data from the entire population



Sampling Effectiveness

- Sampling can help improve effectiveness if information that is more accurate can be obtained
- This is accomplished by talking to fewer employees but asking them questions that are more detailed
- If less people are interviewed, the systems analyst has more time to follow up on missing or incomplete data



Sampling Bias

- Data gathering bias can be reduced by sampling
- When the systems analyst asks for an opinion about a permanent feature of the installed information system, the executive interviewed may provide a biased evaluation because there is little possibility of changing it



Sampling Design

- To design a good sample, a systems analyst must follow four steps:
 - Determining the data to be collected or described
 - Determining the population to be sampled
 - Choosing the type of sample
 - Deciding on the sample size

Four Main Types of Samples the Analyst Has Available (Figure 5.1)

	Not Based on Probability	Based on Probability
Sample elements are selected directly without restrictions	Convenience	Simple random
Sample elements are selected according to specific criteria	Purposive	Complex random (systematic, stratified, and cluster)

The systems analyst should use a complex random sample if possible.



Four Main Types of Samples

- Convenience
- Purposive
- Simple random
- Complex random



Convenience Samples

- Convenience samples are unrestricted, nonprobability samples.
- This sample is the easiest to arrange
- The most unreliable



Purposive Sample

- A purposive sample is based on judgment
- Choose a group of individuals who appear knowledgeable and are interested in the new information system
- A nonprobability sample
- Only moderately reliable



Complex Random Samples

- The complex random samples that are most appropriate for a systems analyst are
 - Systematic sampling
 - Stratified sampling
 - Cluster sampling



The Sample Size Decision

- Determine the attribute
- Locate the database or reports in which the attribute can be found
- Examine the attribute
- Make the subjective decision regarding the acceptable interval estimate
- Choose the confidence level
- Calculate the standard error
- Determine the sample size

A Table of Area under a Normal Curve Can Be Used to Look up a Value Once the Systems Analyst Decides on the Confidence Level (Figure 5.2)

First decide on the confidence level ...

Confidence Level

Confidence Coefficient (z value)

99%

2.58

98

2.33

97

2.17

96

2.05

95

1.96

90

1.65

80

1.28

50

0.67

... then look up the z value.



Calculate the Standard Error of the Proportion

$$\sigma_p = i/z$$

i = interval estimate

z = confidence coefficient found in
the confidence level lookup table

Determine the Sample Size

$$n = \frac{p(1-p)}{\sigma_p^2} + 1$$

σ_p = standard error

p = the proportion of the population having the attribute

Example: A. Sembly Company

- Determine that you are looking for orders with mistakes
- Locate order forms from the past six months
- Examine order forms and conclude that $p = 5\%$
- Subjective decision of acceptable interval $i = \pm 0.02$
- Look up confidence coefficient z - value = 1.96
- Calculate $\sigma_p = i / z = 0.02/1.96 = 0.0102$
- Determine n ; $n = 458$



Investigation

- The act of discovery and analysis of data
- Hard data
 - Quantitative
 - Qualitative



Analyzing Quantitative Documents

- Reports used for decision making
- Performance reports
- Records
- Data capture forms
- Ecommerce and other transactions



Reports Used for Decision Making

- Sales reports
- Production reports
- Summary reports

Performance Report Showing Improvement (Figure 5.3)

Week	Number of Batches Produced	Number of Batches Rejected	Percentage Rejected	Amount Away from 5% Goal
2/2	245	19		
2/9	229	19	7.8	2.8
2/16	219	14	8.3	3.3
2/23	252	13	6.3	1.3
3/2	245	13	5.2	0.2
3/9	260	13	5.3	0.3
3/16	275	14	5.0	***
3/23	260	13	5.1	0.1
3/30	260	13	5.0	***
4/6	244	12	5.0	***
4/13	242	11	4.9	***
4/20	249	11	4.5	***
4/27	249	11	4.4	***
*** indicates met or exceeded the < 5% goal				

Performance
reports
show goals ...

... and trends.





Records

- Records provide periodic updates of what is occurring in the business
- There are several ways to inspect a record:
 - Checking for errors in amounts and totals
 - Looking for opportunities for improving the recording form design
 - Observing the number and type of transactions
 - Watching for instances in which the computer can simplify the work (calculations and other data manipulation)



Data Capture Forms

- Collect examples of all the forms in use
- Note the type of form
- Document the intended distribution pattern
- Compare the intended distribution pattern with who actually receives the form

Questions to Ask about Official and Bootleg Forms that Are Already Filled out (Figure 5.5)

Farmfresh Reorder of Shorted Dairy Products

Date _____ Store Name _____ Store Number _____

Item Requested	Cases	Item Requested	Cases
Milk (1/2 gals.)		Milk (quarts)	
Whole		Whole	
2%		2%	
1%		1%	
Skim		Skim	
Buttermilk		Buttermilk	
Chocolate		Chocolate	
Yogurt			
Plain		Pineapple	
Vanilla		Dutch Apple	
Peach		Banana	
Blueberry		Mixed Fruit	
Boysenberry		Raspberry	
Strawberry		Lemon	
Ice Cream			
Deluxe Pints		Deluxe Quarts	
Deluxe 1/2 Gallons		Premium Pints	
Skinny Minnies		Premium Quarts	

Requested by (employee number) _____ Total Cases Ordered _____
Reason for Shortage _____
Driver Number _____ Route Number _____

Store _____ Date _____ Driver _____

Product shorted _____ Cases needed _____

Dairy manager's initials _____

Official form can overwhelm people by asking for too much information.

There may be no logical order to the form.

Is the total really needed?

"Bootleg" forms arise to simplify the problem.



Questions to Ask About Forms

- Is the form filled out in its entirety?
- Are there forms that are never used?
- Are all copies of forms circulated to the proper people or filed appropriately?
- Can people who must access online forms do so?
- If there is a paper form that is offered as an alternative to a Web-based form, compare the completion rates for both
- Are “unofficial” forms being used on a regular basis?



Analyzing Qualitative Documents

- Key or guiding metaphors
- Insiders vs. outsiders mentality
- What is considered good vs. evil
- Graphics, logos, and icons in common areas or web pages
- A sense of humor



Analyzing Qualitative Documents

- Email messages and memos
- Signs or posters on bulletin boards
- Corporate websites
- Manuals
- Policy handbooks

Analysis of Memos Provides Insight into the Metaphors that Guide the Organization's Thinking (Figure 5.6)

MEMO

To: All Night Call Desk Staff
From: S. Leep, Night Manager
Date: 2/15/2013
Re: Get Acquainted Party Tonight

It's a pleasure to welcome two new 11-7 Call Desk staff members, Twyla Tine and Al Knight. I'm sure they'll enjoy working here. Being together in the wee hours makes us feel like one big happy family. Remember for your breaks tonight that some of the crew has brought in food. Help yourself to the spread you find in the break room, and welcome to the clan, Twyla and Al.



Observation

- Observation provides insight on what organizational members actually do
- See firsthand the relationships that exist between decision makers and other organizational members
- Can also reveal important clues regarding HCI concerns



Analyst's Playscript

- Involves observing the decision-makers behavior and recording their actions using a series of action verbs
- Examples:
 - Talking
 - Sampling
 - Corresponding
 - Deciding

A Sample Page from the Analyst's Playscript (Figure 5.7)

Playscript Analysis Company: Solid Steel Shelving
Analyst: L. Bracket Scenario: Quality Assurance
Date: 1/3/2013

<u>Decision Maker (Actor)</u>	<u>Information-Related Activity (Script)</u>
Quality Assurance Manager	Asks shop floor supervisor for the day's production report
Shop Floor Supervisor	Prints out daily computerized production report
	Discusses recurring problems in production runs with quality assurance (QA) manager
Quality Assurance Manager	Reads production report
	Compares current report with other reports from the same week
	Inputs data from daily production run into QA model on computer
	Observes onscreen results of QA model
	Calls steel suppliers to discuss deviations from quality standards
Shop Floor Supervisor	Attends meeting on new quality specifications with quality assurance manager and vice president of production
Quality Assurance Manager	Drafts letter to inform suppliers on new quality specifications agreed on in meeting
	Sends draft to vice president via email
Vice President of Production	Reads drafted letter
	Returns corrections and comments via email
Quality Assurance Manager	Reads corrected letter on email
	Rewrites letter to reflect changes



STROBE

STRuctured **OB**servation of the **E**nvironment—a technique for observing the decision-maker's physical environment



STROBE

- Often it is possible to observe the particulars of the surroundings that will confirm or negate the organizational narrative
 - Also called stories or dialogue
 - Information that is found through interviews or questionnaires



STROBE Elements

- Office location
- Desk placement
- Stationary equipment
- Props
- External information sources
- Office lighting and color
- Clothing worn by decision makers



Office Location

- Who has the corner office?
- Are the key decision makers dispersed over separate floors?



Desk Placement

- Does the placement of the desk encourage communication?
- Does the placement demonstrate power?



Stationary Office Equipment

- Does the decision maker prefer to gather and store information personally?
- Is the storage area large or small?



Props

- Is there evidence that the decision maker uses a PC, smart phone, or tablet computer in the office?



External Information Sources

- Does the decision maker get much information from external sources such as trade journals or the Web?



Office Lighting and Color

- Is the lighting set up to do detailed work or more appropriate for casual communication?
- Are the colors warm and inviting?



Clothing

- Does the decision maker show authority by wearing conservative suits?
- Are employees required to wear uniforms?

STROBE and Decision-Maker Characteristics (Figure 5.9)

Characteristics of Decision Makers	Corresponding Elements in the Physical Environment
Gathers information informally	Warm, incandescent lighting and colors
Seeks extraorganizational information	Trade journals present in office
Processes data personally	PCs, or tablet computers present in office
Stores information personally	Equipment/files present in office
Exercises power in decision making	Desk placed for power
Exhibits credibility in decision making	Wears authoritative clothing
Shares information with others	Office easily accessible



Applying STROBE

- The five symbols used to evaluate how observation of the elements of STROBE compared with interview results are:
 - A checkmark means the narrative is confirmed
 - An “X” means the narrative is reversed
 - An oval or eye-shaped symbol serves as a cue to look further
 - A square means observation modifies the narrative
 - A circle means narrative is supplemented by observation

An Anecdotal List with Symbols

(Figure 5.10)

Anecdotal List with Symbols for Applying STROBE

Narrative Portrayed by Organization Members	Office Location and Equipment	Office Lighting, Color, and Graphics	Clothing of the Decision Maker
Information is readily flowing on all levels.			
Adams says, "I figure out the percentages myself."			
Vinnie says, "I like to read up on these things."			
Ed says, "The right hand doesn't always know what the left hand is doing."			
Adams says, "Our company doesn't change much."			
The operations staff works all night sometimes.			
Vinnie says, "We do things the way Mr. Adams wants to."			
Julie says, "Stanley doesn't seem to care sometimes."			
			
			
			
			

Key

-  Confirm the narrative
-  Negate or reverse the narrative
-  Cue to look further
-  Modify the narrative
-  Supplement the narrative



Summary

- Sampling
 - Designing a good sample
 - Types of samples
 - Sample size
- Hard data
 - Quantitative document analysis
 - Qualitative document analysis
- Observation
 - Playscript
- STROBE
 - STROBE elements
 - Applying STROBE



END OF
Lecture 05